

SIMATS ENGINEERING
C03 ASSESSMENT TOOL1

COURSE CODE: CSA1610

COURSE NAME: DWDM

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EXPERIMENTS

Q1 R Code for Covariance and Correlation:

CODE:

@relation photo_preference

@attribute Age_of_child {5-6,7-8,9-10}

@attribute Photo_A numeric

@attribute Photo_B numeric

@attribute Photo_C numeric

@data

5-6,18,22,20

7-8,2,28,40

9-10,20,10,40



Q2 R Code for Boxplot and Outlier Detection

CODE:

```
@relation tennis_scores
```

```
@attribute Player_ID string
```

```
@attribute Player_Name string
```

```
@attribute Points_Scored numeric
```

```
@data
```

```
P1,Player_A,12
```

```
P2,Player_B,15
```

```
P3,Player_C,14
```

```
P4,Player_D,10
```

```
P5,Player_E,18
```

```
P6,Player_F,20
```

```
P7,Player_G,22
```

```
P8,Player_H,13
```

```
P9,Player_I,11
```

```
P10,Player_J,35
```

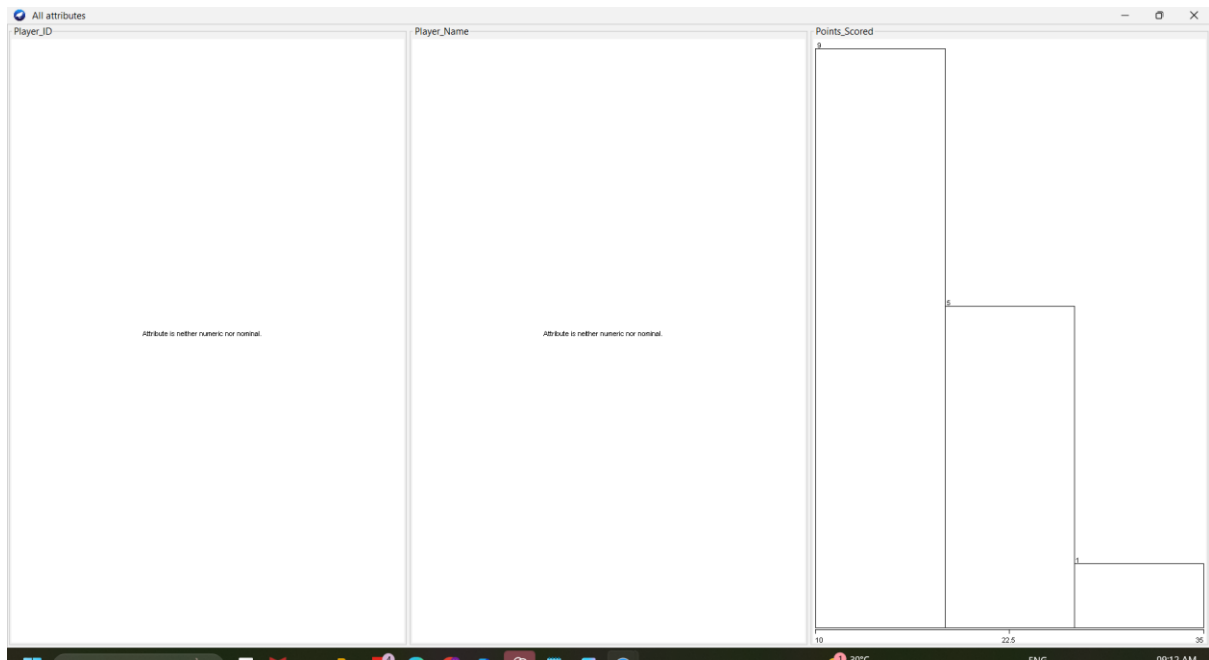
```
P11,Player_K,16
```

```
P12,Player_L,17
```

```
P13,Player_M,19
```

```
P14,Player_N,21
```

```
P15,Player_O,23
```



Q3 R Code for Decision Tree Classifier

CODE:

```
@relation loan_approval
```

```
@attribute Age {young,middle,old}
```

```
@attribute Income {low,medium,high}
```

```
@attribute Employment_Status {employed,unemployed,self-employed}
```

```
@attribute Credit_Score {poor,average,good}
```

```
@attribute Loan_Approval {yes,no}
```

```
@data
```

```
young,high,employed,good,yes
```

```
young,high,self-employed,average,yes
```

```
middle,high,employed,good,yes
```

```
old,medium,employed,good,yes
```

```
old,low,unemployed,poor,no
```

```
old,low,self-employed,average,no
```

```
middle,low,unemployed,poor,no
```

young,medium,employed,average,yes

young,low,unemployed,poor,no

old,medium,self-employed,good,yes

middle,medium,employed,average,yes

middle,high,self-employed,good,yes

old,medium,unemployed,poor,no

young,medium,self-employed,average,yes

middle,low,employed,poor,no

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'J48 -C 0.25 -M 2'. The test options are set to 'Cross-validation' with 10 folds. The target attribute is '(Nom) Loan_Approval'. The classifier output is displayed in the right pane, showing run information and the resulting pruned tree model.

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds: 10
- ☐ Percentage split %: 66

Classifier output:

```
=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    loan_approval
Instances:   15
Attributes:  5
              Age
              Income
              Employment_Status
              Credit_Score
              Loan_Approval
Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
-----

Credit_Score = poor: no (5.0)
Credit_Score = average: yes (5.0/1.0)
Credit_Score = good: yes (5.0)

Number of Leaves :    3

Size of the tree :    4
```

```

Time taken to build model: 0.04 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      12           80      %
Incorrectly Classified Instances    3           20      %
Kappa statistic                    0.5455
Mean absolute error                 0.2767
Root mean squared error            0.4667
Relative absolute error            56.2712 %
Root relative squared error        93.2675 %
Total Number of Instances         15

|
=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                -----  -----  -
                1.000    0.500    0.750     1.000    0.857     0.612    0.611    0.652    yes
                0.500    0.000    1.000     0.500    0.667     0.612    0.611    0.700    no
Weighted Avg.   0.800    0.300    0.850     0.800    0.781     0.612    0.611    0.671

=== Confusion Matrix ===

 a b  <-- classified as
 9 0 | a = yes
 3 3 | b = no

```

Q4 R Code for Mean, Median, Range, and Boxplot

CODE:

```
@relation exam_scores
```

```
@attribute ClassA numeric
```

```
@attribute ClassB numeric
```

```
@data
```

```
76,51
```

```
35,56
```

```
47,84
```

```
64,60
```

```
95,59
```

```
66,70
```

```
89,63
```

```
36,66
```

84,50

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Open file...

Open URL...

Open DB...

Generate...

Undo

Edit...

Save...

Filter

ChooseNone

Apply

Stop

Current relation

Relation: exam_scores

Instances: 9

Attributes: 2

Sum of weights: 9

Attributes

All

None

Invert

Pattern

No.	Name
1	☒ ClassA
2	☐ ClassB

Remove

Selected attribute

Name: ClassA

Missing: 0 (0%)

Distinct: 9

Type: Numeric

Unique: 9 (100%)

Statistic	Value
Minimum	35
Maximum	95
Mean	65.778
StdDev	22.415

Class: ClassB (Num)

Visualize All

4

4.0
[35, 65]

5

35

65

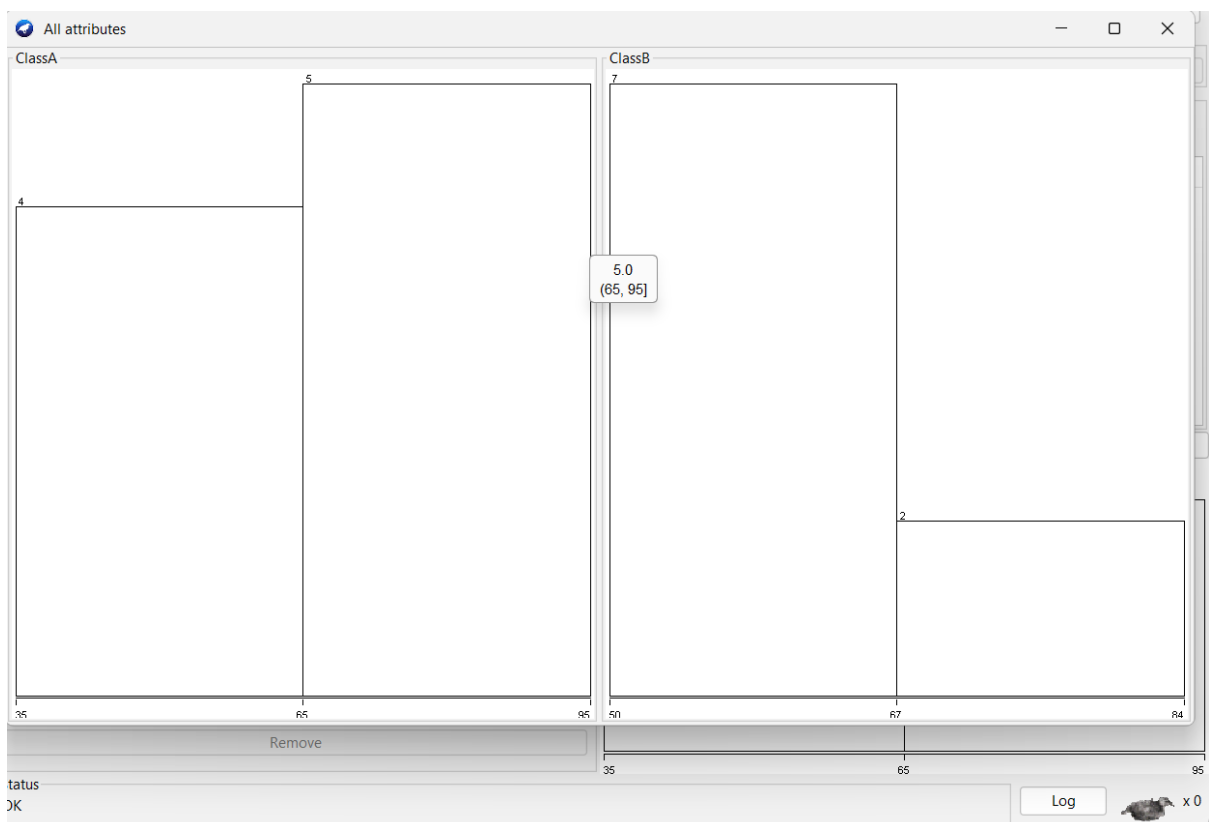
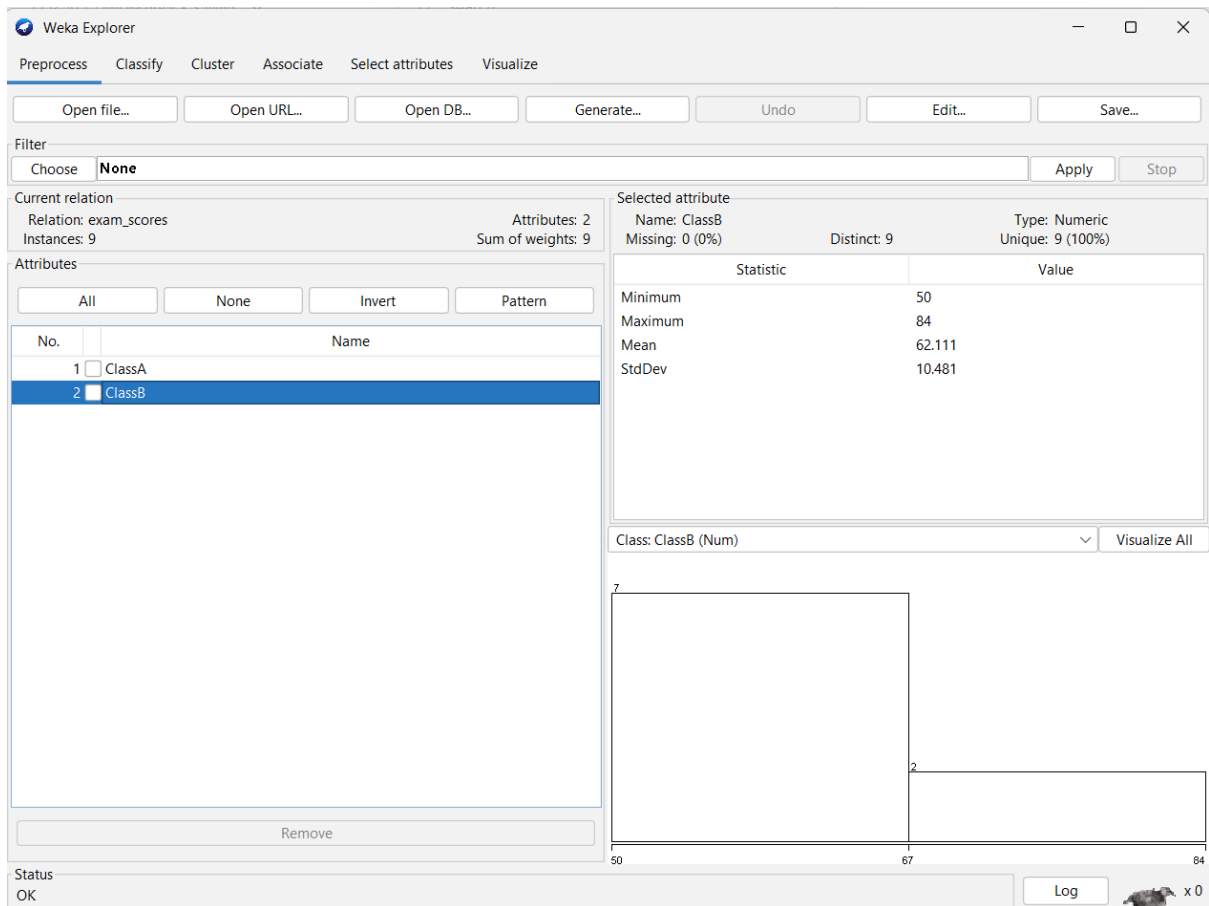
95

Status

OK

Log

 x 0



Q5 R Code for Statistical Analysis

CODE:

```
@relation bodyfat_data
```

```
@attribute age numeric
```

```
@attribute fat numeric
```

```
@data
```

```
23,9.5
```

```
23,27.9
```

```
27,7.8
```

```
37,31.4
```

```
41,25.9
```

```
47,27.4
```

```
49,31.2
```

```
50,31.2
```

```
34,34.6
```

```
42.5,28.8
```

```
38,35.2
```

```
50,32.8
```

```
61,42.5
```

```
45,28.8
```

```
43,35.2
```

```
50,32.8
```

```
61,42.5
```

```
35.7,31.2
```